

Joint Framework of the Estimand Concept and Missing Values in Diagnostic Studies – a Simulation Protocol

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Aims

This simulation study aims to compare the accuracy of different estimand strategies with standard analysis approaches in estimating the sensitivity and specificity of a binary index test with missing values and interfering events in a single-test diagnostic accuracy study.

Data generation

Data for a binary reference standard, a binary index test, three covariates and two interfering events (IE) are generated using a parametric model. The binary reference standard (target condition present/absent) is simulated based on the Binomial distribution $B(n, p)$ in which n is the total sample size of the study and p the prevalence of the target condition. As outlined by Burton et al. (2006), we generate the binary variables (the index test, IE indicator variables, one covariate) as continuous variables and dichotomize them afterwards. Hence, continuous index test values, three continuous covariates, and two continuous IEs are simulated separately for both reference standard groups (target condition present or absent) following a multivariate normal distribution. Both multivariate models differ regarding their mean for the index test (ind), and the IEs (IE1, IE2). The means for the model of the subjects without target condition are set to $\mu = (0, 0, 0, 0, 5, 70)$ for the index test, IE1, IE2 and covariates (z1, z2, z3), respectively. The sample with the target condition has a mean of $\Phi^{-1}(\text{Input Specificity}) - \Phi^{-1}(1 - \text{Input Sensitivity})$ for the index test with Φ^{-1} being the inverse of the cumulative density function of the normal distribution (CDF). With “Input sensitivity” and “input specificity” we refer to the sensitivity and specificity we define as simulation parameter (see Table1) at the start of the data generation. The input sensitivity and specificity do not account for the IEs yet. The means for the IEs in the subjects with the target condition are dependent on the overall proportion of occurring IEs (p_{ie}), the prevalence of the target condition (p) and the difference (effect) between the proportion of occurred IEs in the diseased and in the healthy sample (eff_{ie}), i.e. if $eff_{ie} = 0.1$ then the proportion of occurred IEs in the diseased sample is higher by 0.1 compared to the proportion of occurred IEs in the healthy sample. Based on these parameters the proportion of occurring IEs among the subjects without and with the target condition are calculated by $p_{ie} - eff_{ie}p$ and $p_{ie} - eff_{ie}p + eff_{ie}$, respectively. The mean of the IEs for the diseased sample is then estimated similar to the approach for the index test by $\Phi^{-1}(1 - \text{proportion of occurring IEs in the healthy sample}) - \Phi^{-1}(1 - \text{proportion of occurring IEs in the diseased sample})$. The means for the covariates are the same as for the subjects without the target condition and set to $\mu = (0, 5, 70)$. The data generated under both models are then combined to one dataset and the continuous variables (except z2 and z3) are dichotomized. To maintain the defined input sensitivity and specificity, the index test values are dichotomized at the cut-off value of $\Phi^{-1}(\text{Input specificity})$. One covariate (z1) is dichotomized at its median and the IE variables are dichotomized at $\Phi^{-1}(1 - \text{proportion of occurring IEs in the healthy sample})$. Then, missing values are inserted in the index test variable with the ampute function (van Buuren and Groothuis-Oudshoorn 2011) following the defined missingness mechanism and proportion of missing values as defined in the simulation parameters. In case of MAR, the probability of being missing is dependent on the three covariates, but not the IEs or reference test. Subsequently, the index test values for which an IE 1 is present will be set to positive with a defined probability of pos_{ie} if the IE affects the index test value (determined by

probability $1 - pr_{ind}$). The index test values of subjects who experienced an IE2 (IE2=1) will be set to non-existent (mimicking an IE resulting in a non-existent value). The multivariate models are simulated using the R package mvtnorm (Genz and Bretz 2009) and the missing values are inserted using the R package mice (van Buuren and Groothuis-Oudshoorn 2011). To ensure reproducibility, random seeds are set deterministically as: 5273 for the parallel simulation, $473 * \text{iteration number}$ for the data generation, $56 * \text{iteration number}$ for the multiple imputation. If all subject with the target condition or all subjects without the target condition have missing or non-existent values, the data generation process will be repeated with another seed: $(473 + x) * \text{iteration number}$, with x being the number of repetitions. Computations are run on a high performance cluster with Debian GNU/Linux using parallel processing on 6 nodes with 120 CPU cores each.

Simulation parameter

Table 1 provides an overview of the simulation parameters used for the data generation, which are varied in a fully factorial design. This leads to a total of 768 different scenarios (see the parameter variants for traceability). Each scenario is repeated over 1000 iterations.

Table 1. Overview of simulation parameters.

Simulation parameter	Values	Parameter variants
Input sensitivity (Se)	0.8; 0.9	2
Input specificity (Sp)	0.9; 0.95	2
Sample size (N)	100; 500	2
Prevalence (p)	0.1; 0.3	2
Correlation between index test and covariates and among covariates (r_{cov})	0.4; 0.7	2
Correlation between IEs and covariates ¹ (r_{ie_cov})	0.2	1
Correlation among IEs (r_{ie_ie})	0 (0.2 if both IEs are informative)	1
Correlation between IE1 and index test, and IE2 and the index test (r_{ie_ind}) ¹	0	1
Proportion of IE1 (p_{ie1}) ¹	0.1	1
Difference in the proportion of occurred IE1 between diseased and healthy subjects (eff_{ie1})	0 (non-informative); 0.1 (informative)	2
Probability that the index test is not affected by IE1 (pr_{ind}) ¹	0	1
Probability that the index test is positive if IE1 occurred and affected the index test (pos_{ie1}) ¹	0.8 (for the informative IE); 0.5 (for the non-informative IE)	1
Proportion of IE2 (p_{ie2}) ¹	0.1	1
Difference in the proportion of occurred IE2 between diseased and healthy subjects (eff_{ie2})	0 (non-informative); 0.1 (informative)	2
Proportion of missing values (pm)	0.1; 0.3	2
Missingness mechanism ($mech$)	MCAR; MAR; MNAR ²	3
Missingness pattern ¹	Missing values only in the index test	1
Weights for missingness mechanism ¹	MCAR: Missingness does not depend on any variable MAR: Missingness depends only on the covariates z1-z3	1

	MNAR: Missingness depends only on the index test itself (higher values are more likely to be missing)	
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¹These simulation parameters will not be altered across the scenarios

²MCAR Missing completely at random; MAR Missing at random; MNAR Missing not at random

Estimand/Target of Analysis

The purpose of this simulation study is to compare different estimand (strategies) in combination with different approaches to handle missing values. Thus, the target of the analysis varies by estimand defined. The three estimands differ in their strategies for interfering events as outlined in the following.

Interfering event strategies

Estimand 1: Naive approach

Estimand 1 estimates sensitivity/specificity ignoring IEs and missing data (comparable to a complete case analysis), that is to say using the diagnostic policy strategy (DP) for interfering events and excluding missing values.

Estimand 2: Worst case

Estimand 2 estimates sensitivity/specificity treating IEs as negative results and using single imputation with intention-to-diagnose for missing values. Thus, the setback strategy (SB) is used for IEs.

Estimand 3: Advanced approach

Estimand 3 estimates sensitivity/specificity after imputing missing data with multiple imputation and adjusting for IEs.

- Non-informative IE with an existent index test value: DP
- Informative IE with an existent index test value: principal stratum (PS)
- Non-informative IE with a non-existent index test value: hypothetical (HYP)
- Informative IE with a non-existent index test value: indicative event with imputing a positive index test value (IE+)
- Missing values: multiple imputation (MI), specifically multiple imputation by chained equations (MICE)

Methods

In this section we describe the methods used for each IE strategy defined in the previous section. In addition, we describe the methods used for handling missing values.

Methods for handling IEs in the analysis

The IE strategies are selected in accordance with the defined estimands. For the naive estimand 1 the diagnostic policy strategy is chosen for IE1 (result of the index test is existent) and the principal stratum strategy for IE2 (result of the index test is not existent). The setback strategy is selected for both IEs in the worst case estimand. A mix of the diagnostic policy, principal stratum, hypothetical, and indicative event strategy is chosen to handle the four different IEs in the advanced estimand.

In case of a diagnostic policy strategy the index test result will be used as generated in the dataset. The index test will be integrated in the accuracy calculation without any adjustment. If the IE handled with this strategy is present and the index test result is non-existent, then this result will be handled with the strategy for missing values.

The idea of the principal stratum strategy is to examine the accuracy in the strata with and without the IE. As the number in the strata with the IE is expected to be very small (e.g. in a sample of $N = 100$, $p = 0.1$ and $p_{ie2} = 0.1$ we expect 9 cases in the healthy population and 1 in the diseased population) we will not calculate the sensitivity and specificity. We include only individuals without the specific IE and calculate accuracy based on their available index test result to evaluate the accuracy of the strata without the IE.

The method for the hypothetical strategy is aligned with the method for missing values. A logistic regression will be used for the multiple imputation (for more details see the methods for handling missing values).

We will use a modified single imputation if the IE is handled with the indicative event strategy. If the IE occurred, the index test result will be set to positive nevertheless the index test result is missing, positive or negative.

Mix of strategies

It is possible that both IEs occur in an individual simultaneously. Then we define the result of the index test as non-existent. For the naive approach, we exclude this individual from our analysis as this is not a complete case. For the worst case scenario we impute a false positive or negative result based on the true target condition. For the advanced approach the strategy is dependent on the IEs. If the IE, which leads to an existent result, would be handled with the diagnostic policy strategy, then this is no longer possible as the second IE prevents to observe a test decision. In that case we use either the hypothetical strategy with multiple imputation (in case of non-informative IE) or the indicative event strategy with a single imputation as positive result (in case of an informative IE). Given an IE with an existent result that would be handled with the principal stratum strategy we only analyze the stratum without the event. Therefore, we exclude this case from the analysis. Table 2 shows the different combinations that can happen if both IEs occur in one patient and the respective strategy that will be used in that case.

Table 2. Interfering event strategies for the advanced approach if both IEs occur in one patient

IE1	IE2	Index test value	Strategy
Informative	Informative	Non-existent	PS (this subject is excluded from the analysis)
Informative	Non-informative	Non-existent	PS (this subject is excluded from the analysis)
Non-informative	Informative	Non-existent	IE+
Non-informative	Non-informative	Non-existent	Use HYP (MI)

Methods for handling missing values

The standard complete case analysis excluding all missing values is used as method for handling missing values for estimand 1 (naive approach). The intention-to-diagnose approach (Schuetz et al. 2012) used in estimand 2 (worst case approach) models a worst case scenario by imputing a positive index test value for subjects without the target condition and a negative index test value for subjects with the target condition. In contrast to these simple missing data methods, estimand 3 (advanced approach) employs multiple imputation by chained equations (MICE) to account for MAR and adequate standard errors. As the index test with missing values is binary in our simulation study, we will use logistic regression as imputation method (van Buuren 2018, White et al. 2011). However, perfect prediction can pose a problem (White et al. 2011) when using logistic regression. The number of imputations must be balanced between a high enough number to ensure validity and reproducibility

of imputation results while at the same time small enough so that the computational time of the simulation study does not become inflated. White et al. (2011) affirm the rule of thumb to use as many imputations as the proportion of missing data to ensure appropriate properties of the Monte Carlo error. As the proportion of missing values range from 10% to 30% in our simulation study, this rule would lead to 10 to 30 imputations. To avoid a differing number of imputations throughout our simulation study, we will use 30 imputations for all scenarios. If this leads to an unfeasible high computational time, we will reduce the number of imputations to 20, which is considered sufficient for moderate levels of missingness (van Buuren 2018). The following variables will be included in the imputation model for imputing missing values in the index test: all three covariates (z_1 , z_2 , z_3), and the reference standard. IE1 is only included as predictor variable if IE1 is non-informative since it would be a constant otherwise due to the principal stratum strategy in case of an informative IE1. IE2 is not included in the imputation model as the index test values would always be positive (indicative event strategy) or non-existent (hypothetical strategy) if IE2 is present. Although some diagnostic studies did not include the reference standard in the imputation model, we decided to do so following general recommendations (White et al. 2011, Moons et al. 2006). According to Moons et al. (2006) and White et al. (2011), the outcome – which would correspond to the reference standard in our study – should be included in the imputation model to reduce biasedness. Unfortunately, we have not found any guidance or simulation study on the examination of including or excluding the reference standard to impute the index test in a diagnostic study yet.

Technical implementation

Estimand 1 (naive approach): Sensitivity and specificity and their 95% Wilson score confidence intervals are calculated on the generated data without any further modification of the data.

Estimand 2 (worst case scenario): The index test values will be set to “positive” if IE1 or IE2 are present or the index test value is missing and the reference standard is “healthy”. If the reference standard is “diseased” and IE1 or IE2 are present or the index test value is missing, the index test value will be set to “negative”. Sensitivity and specificity and their 95% Wilson score confidence intervals will be calculated based on the modified dataset.

Estimand 3 (advanced approach): First, if IE1 is informative, the following analysis will be conducted only for subjects with IE1 not present (principal stratum strategy). Otherwise, the whole dataset is used. In both cases, missing values and non-existent results will be imputed by using multiple imputation by chained equations (as described above). If IE2 is informative the imputed index test values for which IE2 is present are manually set to positive following the indicative event strategy. This step cannot be done before multiple imputation as the imputation model would not converge. Sensitivity and specificity estimates and their 95% Wilson score confidence intervals will be calculated based on the imputed datasets. The R package *miceafter* (Heymans 2022) will be used to calculate the Wilson score confidence interval after multiple imputation with the functions `prop_wald` and `pool_prop_wilson`.

Performance Measures

Relative bias and coverage probability will be calculated as main performance parameter. Additionally, absolute bias, root mean squared error (RMSE), and all Monte Carlos standard errors (MCSEs) will be calculated as supplementary measures. As the IEs affect the index test in a meaningful way and each estimand targets a different quantity, the input sensitivity and specificity cannot be used for calculating bias and coverage probability. Instead, we calculated the “true” sensitivity and specificity for each estimand analytically, accounting for the interfering events and their handling strategies (see the Appendix). These calculated sensitivity and specificity are then used to estimate bias, RMSE, and coverage probability, with MCSEs reported alongside. Thus, each estimand has a different “true”

sensitivity and specificity. For illustration purposes, we additionally estimate relative bias using the input sensitivity and specificity and report these in the supplemental material.

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Appendix

Analytical calculation of the Estimands

Se – Sensitivity without IEs (input sensitivity)

Sp – Specificity without IEs (input specificity)

p - Prevalence

p_{ie1} – Probability of occurrence of IE1

p_{ie2} – Probability of occurrence of IE2

eff_{ie1} – Difference in proportion of IE1 between diseased and healthy individuals

pos_{ie1} – Probability of a positive test result after IE1

$p_{ie1/2}$ – Probability that only IE1 is present without IE2

$p_{ie2/1}$ – Probability that only IE2 is present without IE1

p_{ie1U2} – Probability that IE1 and IE2 are present

r_{ie_ie} – Correlation among IEs

se_{str1} – Sensitivity in cases with only IE1

se_{str2} – Sensitivity in cases with only IE2

se_{str3} – Sensitivity in cases with IE1 and IE2

sp_{str1} – Specificity in cases with only IE1

sp_{str2} – Specificity in cases with only IE2

sp_{str3} – Specificity in cases with IE1 and IE2

Estimand 1: Naive approach

$$Se_1 = Se * (1 - ie_{11}) + pos_{ie1} * ie_{11}$$

$$Sp_1 = Sp * (1 - ie_{10}) + (1 - pos_{ie1}) * ie_{10}$$

with

$$ie_{10} = p_{ie1} - eff_{ie1} * p \quad [\text{Probability of an occurring IE1 in healthy individuals}]$$

$$ie_{11} = ie_{10} + eff_{ie1} \quad [\text{Probability of an occurring IE1 in diseased individuals}]$$

Estimand 2: Worst case

$$Se_2 = Se * (1 - p_{ieD})$$

$$Sp_2 = Sp * (1 - p_{ieH})$$

with

$$ie_{20} = p_{ie2} - eff_{ie2} * p \quad [\text{Probability of an occurring IE2 in healthy individuals}]$$

$$ie_{21} = ie_{20} + eff_{ie2} \quad [\text{Probability of an occurring IE2 in diseased individuals}]$$

$$p_{ie1U2_D} = ie_{11} * ie_{21} + r_{ie_ie} * \sqrt{(ie_{11} * (1 - ie_{11}) * ie_{21} * (1 - ie_{21}))}$$

[Probability that IE1 and IE2 are present in diseased individuals]

$$p_{ie1U2_H} = ie_{10} * ie_{20} + r_{ie_ie} * \sqrt{(ie_{10} * (1 - ie_{10}) * ie_{20} * (1 - ie_{20}))}$$

[Probability that IE1 and IE2 are present in healthy individuals]

$$p_{ie1/2_D} = i_{11} - p_{ie1U2_D}$$

[Probability that only IE1 is present without IE2 in diseased individuals]

$$p_{ie2/1_D} = i_{21} - p_{ie1U2_D}$$

[Probability that only IE2 is present without IE1 in diseased individuals]

$$p_{ie1/2_H} = i_{10} - p_{ie1U2_H}$$

[Probability that only IE1 is present without IE2 in healthy individuals]

$$p_{ie2/1_H} = i_{20} - p_{ie1U2_H}$$

[Probability that only IE2 is present without IE1 in healthy individuals]

$$p_{ieD} = P_{ie1U2_D} + P_{ie1/2_D} + P_{ie2/1_D}$$

[Probability of at least one IE in diseased individuals]

$$p_{ieH} = P_{ie1U2_H} + P_{ie1/2_H} + P_{ie2/1_H}$$

[Probability of at least one IE in healthy individuals]

Estimand 3: Advanced approach

General formula:

$$Se_3 = Se * (1 - p_{ieD}) + se_{str1} * p_{ie1/2_D} + se_{str2} * p_{ie2/1_D} + se_{str3} * p_{ie1U2_D}$$

$$Sp_3 = Sp * (1 - p_{ieH}) + sp_{str1} * p_{ie1/2_H} + sp_{str2} * p_{ie2/1_H} + sp_{str3} * p_{ie1U2_H}$$

3a) IE1 informative and IE2 non-informative (PS + HYP)

$$Se_3 = Se$$

$$Sp_3 = Sp$$

3b) IE1 non-informative and IE2 informative (DP + IE+)

$$Se_3 = Se * (1 - p_{ieD}) + pos_{ie1} * p_{ie1/2_D} + ie_{21}$$

$$Sp_3 = Sp * (1 - p_{ieH}) + (1 - pos_{ie1}) * p_{ie1/2_H}$$

3c) IE1 and IE2 non-informative (DP + HYP)

$$Se_3 = Se * (1 - p_{ieD} + ie_{21}) + pos_{ie1} * p_{ie1/2_D}$$

$$Sp_3 = Sp * (1 - p_{ieH} + ie_{20}) + (1 - pos_{ie1}) * p_{ie1/2_H}$$

3d) IE1 and IE2 informative (PS + IE+)

$$Se_3 = Se * \frac{1 - p_{ieD}}{1 - ie_{11}} + \frac{ie_{21} - p_{ieD}}{1 - ie_{11}}$$

$$Sp_3 = Sp * \frac{1 - p_{ieH}}{1 - ie_{10}}$$